CLASSWORK 9

Code Comprehension on Regression

CPE 486/586

Instructor: Rahul Bhadani **Due: October 14, 2026, In-Class**

20 points

Consider

```python
import torch
import torch.nn as nn
import torch.optim as optim
class RegressionModel:

    def __init__(self, learning_rate: float = 0.1, max_epochs: int = 1000,
                 tolerance: float = 1e-6):
        self.learning_rate = learning_rate
        self.max_epochs = max_epochs
        self.tolerance = tolerance
        self.w_0 = nn.Parameter(torch.randn(1))
        self.w_1 = nn.Parameter(torch.randn(1))
        self.w_2 = nn.Parameter(torch.randn(1))
        self.optimizer = optim.SGD([self.w_0, self.w_1, self.w_2], lr=self.learning_rate)
        self.loss_history = []

    def forward(self, X1: torch.Tensor, X2: torch.Tensor) -> torch.Tensor:
        return self.w_1 * X1 ** 2 + self.w_2 * torch.log(X2) + self.w_0

    def criterion(self, y_pred: torch.Tensor, y: torch.Tensor) -> torch.Tensor:
        return torch.mean(torch.abs(y_pred - y))

    def fit(self, X1: torch.Tensor, X2: torch.Tensor, y: torch.Tensor):
        previous_loss = float("inf")
        for epoch in range(self.max_epochs):
            self.optimizer.zero_grad()
            y_pred = self.forward(X1, X2)
            loss = self.criterion(y_pred, y)
            loss.backward()
            self.optimizer.step()
            self.loss_history.append(loss.item())
            if abs(previous_loss - loss.item()) < self.tolerance:
                break
            previous_loss = loss.item()
        return self

    def predict(self, X1: torch.Tensor, X2: torch.Tensor) -> torch.Tensor:
        with torch.no_grad():
            return self.forward(X1, X2)
```

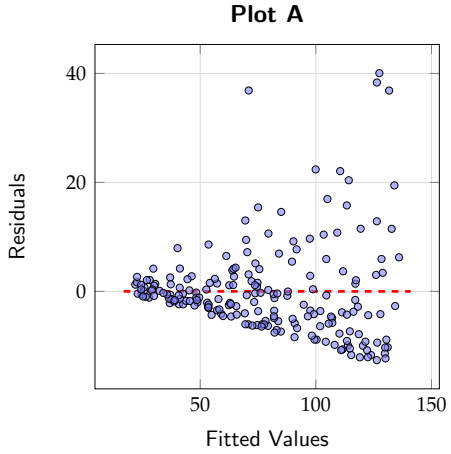
1. Write the expression for identified regression model from the code. **(5 Points)**

Answer

2. Write the expression for identified loss function from the code.**(5 Points)**

Answer

3. Consider the plot shown below and choose the appropriate response: **(5 Points)**



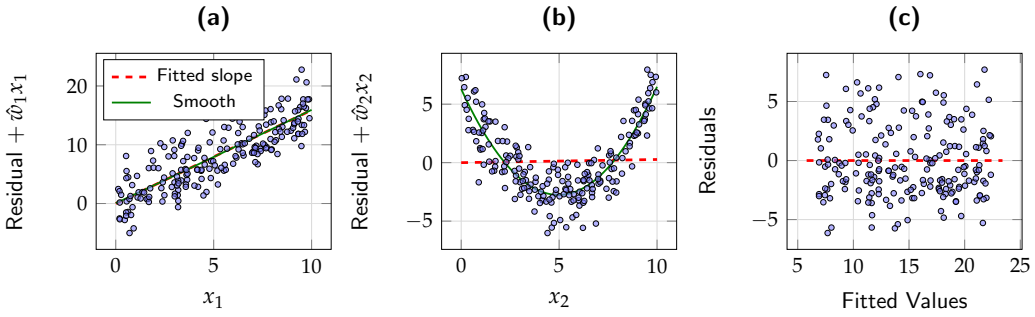
(a) It satisfies constant error variance assumption of the linear regression model.

(b) Error variance grows with the fitted values and hence it exhibits heteroscedasticity.

(c) Error variance decays with the fitted values and hence it exhibits homoscedasticity.

(d) It exhibits the presence of multicollinearity in the data.

4. Residual diagnostics for the model $y \sim x_1 + x_2$ are shown. Panels (a) and (b) are partial residual plots for x_1 and x_2, and panel (c) plots the residuals against the fitted values. Clearly,there is something wrong with the model. How would you fix it? **(5 Points)**



(a) Remove x_2 from the model, since its fitted slope in panel (b) is close to zero.

(b) Add a quadratic term in x_2, giving $y \sim x_1 + x_2 + x_2^2$.

(a) Apply a log transformation to y to stabilize the error variance.

(b) Add a quadratic term in x_1, giving $y \sim x_1 + x_1^2 + x_2$.